

ANSWERS OF MODEL TEST PAPER 8
INTERMEDIATE: GROUP – II
PAPER – 6: FINANCIAL MANAGEMENT & STRATEGIC MANAGEMENT
PAPER 6A : FINANCIAL MANAGEMENT

Suggested Answers/ Hints

PART I – Case Scenario based MCQs

1. i. (D) ₹ 3.3779
- ii. (B) ₹ 8.3655
- iii. (A) ₹ 72.28
- iv. (C) ₹ 45.79
- v. (B) ₹ 54.33

Intrinsic Value = Sum of PV of Expected Dividends + PV of Share Price at the end of the period

The following steps are required:

- A. Determine PV of expected dividends to be received in the next four years.
- B. Determine PV share at the end of 4th Year.
- C. Add the values of A and B above.

(A)

Year	$D_1 = D_0(1+g)$	PV Discount Factor @ 12%	PV in ₹
1	$2(1+14\%) = 2.28$	0.893	2.0364
2	$2.28(1+14\%) = 2.5992$	0.797	2.0715
3	$2.5992(1+14\%) = 2.9631$	0.712	2.1097
4	$2.9631(1+14\%) = 3.3779$	0.636	2.1483
(A) Total PV of Expected Dividend			₹ 8.3655

$$P_4 = \frac{D_5}{K_e - g} = \frac{D_4(1+g)}{K_e - g} = \frac{3.3779(1+7\%)}{12\% - 7\%} = ₹ 72.28$$

(B) PV of share at the end of 4th Year = ₹ 72.28 x 0.636 = ₹ 45.97

(C) Market Price of shares = ₹ 8.3655 + ₹ 45.97 = ₹ 54.33

2. (B) 6.16%

To calculate WACC, we use the formula:

$$WACC = (E/V) \times Re + (D/V) \times Rd \times (1 - T_c)$$

Let V be the total value of the firm, then Debt is equal to 1.5/(1+1.5) times the value of the firm and Equity is equal to 1/(1+1.5) times the value of the firm.

So, $D/V = 1.5/(1+1.5) = 0.6$ and $E/V = 1/(1+1.5) = 0.4$

WACC = $0.4 \times 10\% + 0.6 \times 6\% \times (1 - 40\%) = 4\% + 2.16\% = 6.16\%$
 Therefore, the company's weighted average cost of capital is 6.16%.

3. (A) 1.11

$$\text{EBIT} = 3,00,000 \times (3-1) - 3,50,000 = 2,50,000,$$

$$\text{PBT} = 2,50,000 - 25,000 - 2,25,000$$

$$\text{FL} = 2,50,000/2,25,000 = 1.11$$

4. (C) both automatic and approval route

PART II – Descriptive Questions

1. (a) Calculation of Cost of Preference Shares (K_p)

$$\text{Preference Dividend (PD)} = 0.12 \times 40,000 \times 100 = 4,80,000$$

$$\text{Floatation Cost} = 40,000 \times 2 = ₹ 80,000$$

$$\text{Net Proceeds (NP)} = 42,00,000 - 80,000 = 41,20,000$$

$$\text{Redemption Value (RV)} = 40,000 \times 110 = 44,00,000$$

$$\text{Cost of Redeemable Preference Shares} = \frac{\text{PD} + (\text{RV} - \text{NP}) / N}{\frac{\text{RV} + \text{NP}}{2}}$$

$$\begin{aligned} K_p &= \frac{4,80,000 + (44,00,000 - 41,20,000) / 10}{\frac{44,00,000 + 41,20,000}{2}} \\ &= \frac{4,80,000 + (2,80,000) / 10}{85,20,000 / 2} \\ &= \frac{4,80,000 + 28,000}{42,60,000} = \frac{5,08,000}{42,60,000} \\ &= 0.1192 \\ K_p &= 11.92\% \end{aligned}$$

(Note: K_p may be computed alternatively by taking the RV and NP for one unit of preference shares. Final figure would remain unchanged).

(b) Calculation of Net Cash flow

$$\text{Contribution} = (3.00 - 1.75) \times 50,000 = ₹ 62,500$$

$$\text{Fixed costs} = 40,000 - [(1,25,000 - 30,000)/5] = ₹ 21,000$$

Year	Capital (₹)	Contribution (₹)	Fixed costs (₹)	Adverts (₹)	Net cash flow (₹)
0	(1,00,000)	-	-	-	(1,00,000)
1	(25,000)	62,500	(21,000)	(10,000)	6,500
2	-	62,500	(21,000)	(15,000)	26,500
3	-	62,500	(21,000)	-	41,500

4	-	62,500	(21,000)	-	41,500
5	30,000	62,500	(21,000)	-	71,500

Calculation of Net Present Value

Year	Net cash flow (₹)	10% discount factor	Present value (₹)
0	(1,00,000)	1.000	(1,00,000)
1	6,500	0.909	5,909
2	26,500	0.826	21,889
3	41,500	0.751	31,167
4	41,500	0.683	28,345
5	71,500	0.621	44,402
NPV			31,712

The net present value of the project is ₹ 31,712.

(c)

	(₹)
Sales	24,00,000
Less: Variable cost	12,00,000
Contribution	12,00,000
Less: Fixed cost	10,00,000
EBIT	2,00,000
Less: Interest	1,00,000
EBT	1,00,000
Less: Tax (50%)	50,000
EAT	50,000
No. of equity shares	10,000
EPS	5

$$(a) \text{ Operating Leverage} = \frac{\text{₹}12,00,000}{\text{₹}2,00,000} = 6 \text{ times}$$

$$(b) \text{ Financial Leverage} = \frac{\text{₹}2,00,000}{\text{₹}1,00,000} = 2 \text{ times}$$

$$(c) \text{ Combined Leverage} = \text{OL} \times \text{FL} = 6 \times 2 = 12 \text{ times.}$$

$$(d) \text{ ROI} = \frac{\text{₹}50,000}{\text{₹}10,00,000} \times 100 = 5\%$$

Here ROI is calculated as ROE i.e. $\frac{\text{EAT} - \text{Pref.Dividend}}{\text{Equity shareholders' fund}}$

$$(e) \text{ Operating Leverage} = 6$$

$$6 = \frac{\Delta \text{ EBIT}}{0.25}$$

$$\Delta \text{ EBIT} = \frac{6 \times 1}{4} = 1.5$$

$$\text{Increase in EBIT} = ₹ 2,00,000 \times 1.5$$

$$= ₹ 3,00,000$$

$$\text{New EBIT} = ₹ 5,00,000$$

2. Working Notes:

1. Raw Material Storage Period (R)

$$\begin{aligned} &= \frac{\text{Average Stock of Raw Material}}{\text{Annual Consumption of Raw Material}} \times 365 \\ &= \frac{\frac{₹ 45,000 + ₹ 65,356}{2}}{₹ 3,79,644} \times 365 \\ &= 53 \text{ days.} \end{aligned}$$

Annual Consumption of Raw Material = Opening Stock + Purchases - Closing Stock

$$= ₹ 45,000 + ₹ 4,00,000 - ₹ 65,356$$

$$= ₹ 3,79,644$$

2. Work-in-Progress (WIP) Conversion Period (W)

$$\begin{aligned} \text{WIP Conversion Period} &= \frac{\text{Average Stock of WIP}}{\text{Annual Cost of Production}} \times 365 \\ &= \frac{\frac{₹ 35,000 + ₹ 51,300}{2}}{₹ 7,50,000} \times 365 \\ &= 21 \text{ days} \end{aligned}$$

3. Finished Stock Storage Period (F)

$$\begin{aligned} &= \frac{\text{Average Stock of Finished Goods}}{\text{Cost of Goods Sold}} \times 365 \\ &= \frac{₹ 65,178}{₹ 9,15,000} \times 365 = 26 \text{ days.} \end{aligned}$$

$$\begin{aligned} \text{Average Stock} &= \frac{₹ 60,181 + ₹ 70,175}{2} \\ &= ₹ 65,178. \end{aligned}$$

4. Debtors Collection Period (D)

$$\begin{aligned} &= \frac{\text{Average Debtors}}{\text{Annual Credit Sales}} \times 365 \\ &= \frac{₹ 1,23,561.50}{₹ 11,00,000} \times 365 \end{aligned}$$

= 41 days

$$\text{Average debtors} = \frac{\text{₹1,12,123} + \text{₹1,35,000}}{2} = \text{₹1,23,561.50}$$

5. **Creditors Payment Period (C)**

$$= \frac{\text{Average Creditors}}{\text{Annual Net Credit Purchases}} \times 365$$

$$= \frac{\left(\frac{\text{₹50,079} + \text{₹70,469}}{2} \right)}{\text{₹4,00,000}} \times 365$$

= 55 days

(i) **Operating Cycle Period**

$$= R + W + F + D - C$$

$$= 53 + 21 + 26 + 41 - 55$$

= 86 days

(ii) **Number of Operating Cycles in the Year**

$$= \frac{365}{\text{Operating Cycle Period}} = \frac{365}{86} = 4.244$$

(iii) **Amount of Working Capital Required**

$$= \frac{\text{Annual Operating Cost}}{\text{Number of Operating Cycles}} = \frac{\text{₹ 9,50,000}}{4.244} = \text{₹ 2, 23,845.42}$$

3. (a) Plan I = Raising Debt of ₹ 2.5 lakh + Equity of ₹ 22.5 lakh
 Plan II = Raising Debt of ₹ 10 lakh + Equity of ₹ 15 lakh
 Plan III = Raising Debt of ₹ 15 lakh + Equity of ₹ 10 lakh

Calculation of Earnings per share (EPS):

Particulars	FINANCIAL PLANS		
	Plan I	Plan II	Plan III
	₹	₹	₹
Expected EBIT	5,00,000	5,00,000	5,00,000
Less: Interest ^(a)	(25,000)	(1,37,500)	(2,37,500)
Earnings before taxes	4,75,000	3,62,500	2,62,500
Less: Taxes @ 50%	(2,37,500)	(1,81,250)	(1,31,250)
Earnings after taxes (EAT)	2,37,500	1,81,250	1,31,250
Number of shares ^(b)	15,000	10,000	8,000
Earnings per share (EPS)	15.83	18.13	16.41

Financing Plan II (i.e. Raising debt of ₹ 10 lakh and issue of equity share capital of ₹ 15 lakh) is the option which maximises the earnings per share.

Working Notes:

(a) Calculation of interest on Debt

Plan		₹	₹
I	(₹ 2,50,000 ´ 10%)		25,000
II	(₹ 2,50,000 ´ 10%)	25,000	
	(₹ 7,50,000 ´ 15%)	1,12,500	1,37,500
III	(₹ 2,50,000 ´ 10%)	25,000	
	(₹ 7,50,000 ´ 15%)	1,12,500	
	(₹ 5,00,000 ´ 20%)	1,00,000	2,37,500

(b) Number of equity shares to be issued

$$\text{Plan I} = \frac{\text{₹ 22,50,000}}{\text{₹ 150 (Market price of share)}} = 15,000 \text{ shares}$$

$$\text{Plan II} = \frac{\text{₹ 15,00,000}}{\text{₹ 150}} = 10,000 \text{ shares}$$

$$\text{Plan III} = \frac{\text{₹ 10,00,000}}{\text{₹ 125}} = 8,000 \text{ shares}$$

(b)

Ratios	Navya Ltd.	Industry Norms
1. Current Ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}}$	$\frac{\text{₹52,80,000}}{\text{₹19,80,000}} = 2.67$	2.50
2. Receivable Turnover Ratio = $\frac{\text{Sales}}{\text{Debtors}}$	$\frac{\text{₹1,10,00,000}}{\text{₹11,00,000}} = 10.0$	8.00
3. Inventory turnover ratio = $\frac{\text{Sales}}{\text{Stock}}$	$\frac{\text{₹1,10,00,000}}{\text{₹33,00,000}} = 3.33$	9.00
4. Total Asset Turnover ratio = $\frac{\text{Sales}}{\text{Total Assets}}$	$\frac{\text{₹1,10,00,000}}{\text{₹77,00,000}} = 1.43$	2.00
5. Net Profit Ratio = $\frac{\text{Net Profit}}{\text{Sales}}$	$\frac{\text{₹2,31,000}}{\text{₹1,10,00,000}} = 2.10\%$	3.50%
6. Return on Total Asset = $\frac{\text{EBIT}}{\text{Total Assets}}$	$\frac{\text{₹5,54,000}}{\text{₹77,00,000}} = 7.19\%$	7%

7. Return on Net worth = $\frac{\text{Net Profit}}{\text{Net Worth}}$	$\frac{₹2,31,000}{₹48,00,000} = 4.81\%$	10.5%
8. $\frac{\text{Total Debt}}{\text{Total Assets}}$	$\frac{₹29,00,000}{₹77,00,000} = 37.66\%$	60%

Comments:

1. The position of Navya Ltd. is better than the industry norm with respect to Current Ratio and Receivables Turnover Ratio.
2. However, the Inventory turnover ratio and Total Asset Turnover ratio is poor comparing to industry norm indicating that company is inefficient to utilize its inventory and assets.
3. The firm also has its net profit ratio and return on net worth ratio much lower than the industry norm.
4. Total debt to total assets ratio is lower than the industry standard which suggests that the firm is less levered by debt and more by equity resulting in less risky company.

4. (a) **Inter-relationship between Investment, Financing and Dividend Decisions:** The finance functions are divided into three major decisions, viz., investment, financing and dividend decisions. It is correct to say that these decisions are inter-related because the underlying objective of these three decisions is the same, i.e. maximisation of shareholders' wealth. Since investment, financing and dividend decisions are all interrelated, one has to consider the joint impact of these decisions on the market price of the company's shares and these decisions should also be solved jointly. The decision to invest in a new project needs the finance for the investment. The financing decision, in turn, is influenced by and influences dividend decision because retained earnings used in internal financing deprive shareholders of their dividends. An efficient financial management can ensure optimal joint decisions. This is possible by evaluating each decision in relation to its effect on the shareholders' wealth.

The above three decisions are briefly examined below in the light of their inter-relationship and to see how they can help in maximising the shareholders' wealth i.e. market price of the company's shares.

Investment decision: The investment of long term funds is made after a careful assessment of the various projects through capital budgeting and uncertainty analysis. However, only that investment proposal is to be accepted which is expected to yield at least so much return as is adequate to meet its cost of financing. This has an influence on the profitability of the company and ultimately on its wealth.

Financing decision: Funds can be raised from various sources. Each source of funds involves different issues. The finance manager has to maintain a proper balance between long-term and short-term funds. With the total volume of long-term funds, he has to ensure a proper mix of

loan funds and owner's funds. The optimum financing mix will increase return to equity shareholders and thus maximise their wealth.

Dividend decision: The finance manager is also concerned with the decision to pay or declare dividend. He assists the top management in deciding as to what portion of the profit should be paid to the shareholders by way of dividends and what portion should be retained in the business. An optimal dividend pay-out ratio maximises shareholders' wealth.

The above discussion makes it clear that investment, financing and dividend decisions are interrelated and are to be taken jointly keeping in view their joint effect on the shareholders' wealth.

- (b) The financing of current assets involves a trade off between risk and return. A firm can choose from short or long term sources of finance. Short term financing is less expensive than long term financing but at the same time, short term financing involves greater risk than long term financing.

Depending on the mix of short term and long term financing, the approach followed by a company may be referred as matching approach, conservative approach and aggressive approach.

In matching approach, long-term finance is used to finance fixed assets and permanent current assets and short term financing to finance temporary or variable current assets. Under the conservative plan, the firm finances its permanent assets and also a part of temporary current assets with long term financing and hence less risk of facing the problem of shortage of funds.

An aggressive policy is said to be followed by the firm when it uses more short term financing than warranted by the matching plan and finances a part of its permanent current assets with short term financing.

- (c) **Optimum Capital Structure:** The capital structure is said to be optimum when the firm has selected such a combination of equity and debt so that the wealth of firm is maximum. At this capital structure, the cost of capital is minimum and the market price per share is maximum.

Or

- (c) In dividend price approach, cost of equity capital is computed by dividing the current dividend by average market price per share. This ratio expresses the cost of equity capital in relation to what yield the company should pay to attract investors. It is computed as:

$$K_e = \frac{D_1}{P_0}$$

Where,

D_1 = Dividend per share in period 1

P_0 = Market price per share today